

EEG Feature Extraction and Pattern Recognition During Right and Left Hands Motor Imagery in Brain-Computer Interface

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Abstract—at present, it is an important issue to obtain accurate classification of EEG signal in brain-computer interface. We collected the EEG data which are recorded at a sampling rate of 250Hz by EGI-64 scalp electrodes placed according to the international 10/20 system. Firstly, discrete wavelet transform method is used to decompose the average power of the channel C3/C4 and P3/P4 in left and right hands imagine movement during the some experimental period of time. The reconstructed signal of approximation coefficient A6 on the sixth level is selected to build up a feature signal. Secondly, the performances by Fisher Linear Discriminant Analysis and Support Vector Machine methods are compared. The final classifications results show that correct classification rate by Support Vector Machine is higher and gain an ideal classification results, which can establish a basis of real-time BCI application.

Keywords—Brain-Computer Interface; Feature Extraction; Motor Imagery; Pattern Classification

I. INTRODUCTION

There are some kinds of diseases which can damage the neuromuscular communication between brain and the outside environment, such as amyotrophic lateral sclerosis (ALS) [1]. These diseases cause people to lose their movement ability gradually and at a later stage people may become totally paralyzed. Although modern life-support technology can prolong patients' life, they live with low quality of life and bring their families and society heavy burden. People try to build a kind of communication and control channel directly between the brain and external devices, without the participation of peripheral nerves and muscles, which named Brain Computer Interface (BCI) [2].

A number of groups around the world are developing BCI systems, in which surface electroencephalography (EEG) is used because it is noninvasive and mobile. The cognitive task most commonly used in BCI studies is motor imagery, as it produces changes in EEG that occur naturally in movement planning and are relatively straightforward to detect. Motor imagery, defined as mental simulation of a kinesthetic movement [3], has been well established that the imagination of each left and right hand movement results in event-related desynchronization (ERD) of mu-band power in the contralateral sensorimotor areas, which is also the case for

physical hand movements [4]. Brain activities modulated by motor imagery of either the left or right hand are regarded as good features for BCIs, because such activities are readily reproducible and show consistent EEG patterns on the sensorimotor cortical areas [5]. Moreover, thanks to the contralateral localization of the oscillatory activity, the activities evoked from left and right hand motor imagery are, comparatively, readily discriminated [6]. The signals generated in the motor cortex can be recorded from electrodes over central head regions and the research has produced encouraging results [7][8][9]. The aim is to train subjects to control the cursor automatically by using the operant conditioning approach.

Effective feature extraction method is an important approach to improve the recognition accuracy. There are several feature extraction methods, such as: (1) Power Spectral Analysis. Power Spectral Analysis can reflect variances of signals' power, but the time information may be lost. (2) Adaptive Autoregressive (AAR). AAR model can reflect variances of event-related electroencephalograph (EEG) without prior information of related bands and is suitable for online analysis. However, artifacts should be controlled effectively in application of this method. (3) Common Spatial Pattern (CSP) [10]. The goal of CSP is to design spatial filters which lead to new signal matrix whose variances are optimal for the discrimination of two classes of EEGs. Specific frequency bands need not to be pre-selected, but multi-electrode are required for the EEG analysis which limits the application of CSP in portable BCI system. So it is necessary to optimize the number of electrodes further. As a classifier, linear discriminant analysis (LDA), support vector machine (SVM) and neural network are used [11].

In this paper, we propose a method to classify the EEG of mental tasks for left-hand movement imagination and right-hand movement imagination [12]. Comparing with what we study before, the difference is that the data in this paper is real EEG data which we collect in our laboratory, and we also compare and analyse the data of P3 and P4 electrode. The goal of this study is to prove the viability of a movement imagery prediction brain-computer interface (BCI), and to expect that the two mental tasks can be classified with higher classification rate. Firstly, a rational time window is set by calculating the

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average power of channel C3/C4 and P3/P4 in left and right hands imagine movement. The time window's average power is then decomposed by discrete wavelet transform (DWT). The reconstructed signal of approximation coefficient A6 on the sixth level is selected to build up a feature signal. Secondly, Fisher Linear Discriminant Analysis (FLDA) is used with two different threshold calculation ways and obtain good misclassification rate. We also use Support Vector Machine (SVM) to compare the performance with Fisher Linear Discriminant Analysis. The final classification results show that correct classification rate by Support Vector Machine is higher and gain an ideal classification results.

II. METHOD

In this paper, we use data which is collected during the BCI experiments, and analyze them after the experiments, we would explain it later in this part.

A. Data Set Description

This dataset is recorded from three normal subjects (2 males, 24y) during a feedback session. The subject sat in a relaxing chair with armrests. The task is to control a feedback bar by means of imagery left or right hand movements. The order of left and right cues is random. The experiment consists of 7 runs with 40 trials each. All runs are conducted on the same day with several minutes break in between. Given are 280 trials of 9s length. The first 2s is quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial with the "beep", and a cross "+" is displayed for 1s; then at $t=3s$, an arrow (left or right) is displayed as cue. At the same time the subject is asked to move a bar into the direction of the cue. The EEG data are recorded at a sampling rate of 250Hz by EGI-64 scalp electrodes and defined by 10-20 system. It is filtered between 0.01Hz and 40Hz. The figure 1 shows the experiment paradigm.

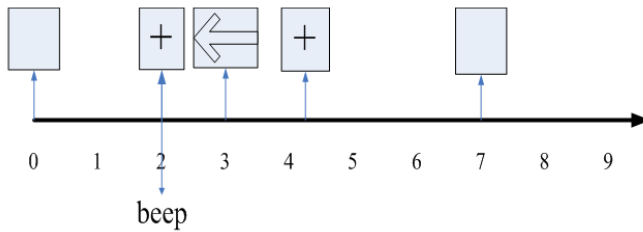


Figure 1. Timing scheme

B. Preprocessing

In this paper, we use Net-Station software to preprocess the data collected from our laboratory, and the steps are Filtering, Lowpass filtering, Segmentation, Artifact Detection, Bad Channel Replacement, Average, Montage Operations, Baseline Correction. In the next step, we use the process result with Baseline Correction for the feature extraction.

C. Feature Extraction

Because the channel Cz show its independence of the motor imagery, channels C3, C4, P3, P4 are chosen to used for feature extraction. The duration of each experiment is 9 seconds, but

the part which played a role for the task of classification may be just one period. In order to find a suitable time window, the average power of different imagine tasks of C3 and C4 channel is computed firstly, so as to provide effective reference of the next step. And the process steps in channel P3/P4 are the same as the process steps in channel C3/C4. Here, we just display some results of the process steps in channel C3/C4.

The formula for calculating average power is:

$$P_{(j)} = \frac{1}{N} \sum_{i=1}^N x_{f(i,j)}^2 \quad (1)$$

$x_{f(i,j)}^2$ is the j th EEG data of the i th time experiment, $P(j)$ is the average power of the j th EEG data corresponding to the N time experiments.

We could calculate the average power of C3 and C4 when one imagined left-hand movement and one imagined right-hand movement shown in Figure 2, 3. From these two figures, we can see that the average power of the C3 electrode is stronger than the average power of the C4 electrode in 0.6s to 0.8s, obviously.

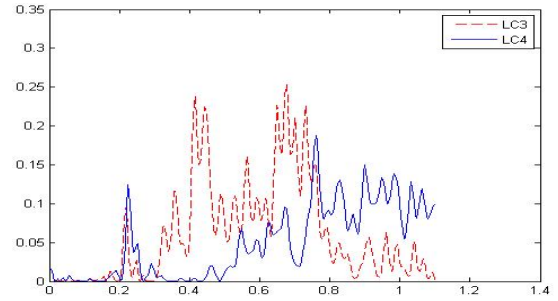


Figure 2. The relationship between the time and the average power of the C3 electrode and C4 electrode for left-hand imagery movement

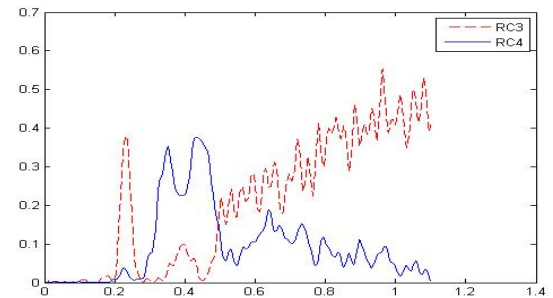


Figure 3. The relationship between the time and the average power of the C3 electrode and C4 electrode for right-hand imagery movement

In this study, we choose the Daubechies wavelets as the basic wavelet functions for their simplicity and general purpose applicability in a variety of time-frequency representation problems. According to the sampling frequency of 250Hz, a 6 level decomposition to the average power at time window is used, thus having 6 scales of details (d1-d6) and a final approximation (a6). The decomposition results are shown in Figures 4, 5.

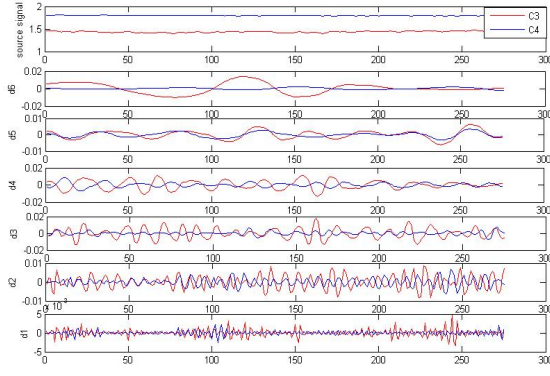


Figure 4. The wavelet decomposition of the C3 electrode and C4 electrode for left-hand imagery movement

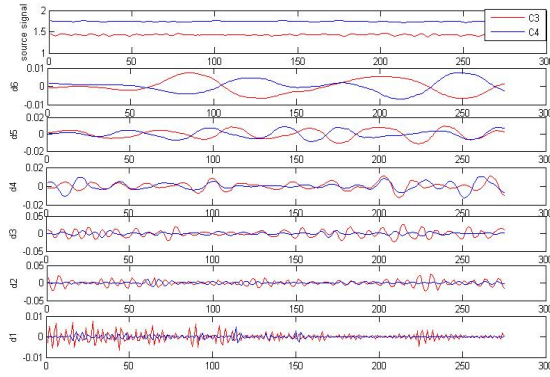


Figure 5. The wavelet decomposition of the C3 electrode and C4 electrode for right-hand imagery movement

Here, the final approximation signals of C3 and C4 when imagined left-hand movement are indicated as LC3A6 and LC4A6, respectively, while the final approximation signals of C3 and C4 when imagined right-hand movement are indicated as RC3A6 and RC4A6, respectively. The results show that there is significant difference between left-hand and right-hand imagery movement concerning the signal of approximation coefficient A6 on the sixth level. The value of LC3A6 is bigger than LC4A6 when imagined left-hand movement, while the value of RC3A6 is smaller than RC4A6 when imagined right-hand movement. Supposing LA6 and RA6 are the difference of the signal of approximation coefficient A6 on the sixth level of C3 and C4, respectively, they are defined as the following equations:

$$LA6 = LC3A6 - LC4A6 \quad (2)$$

$$RA6 = RC3A6 - RC4A6 \quad (3)$$

The curve of LA6 and RA6 is shown in Figure 6. Obviously, LA6 and RA6 can be selected as the features for distinguish the left-right hands imagery movement.

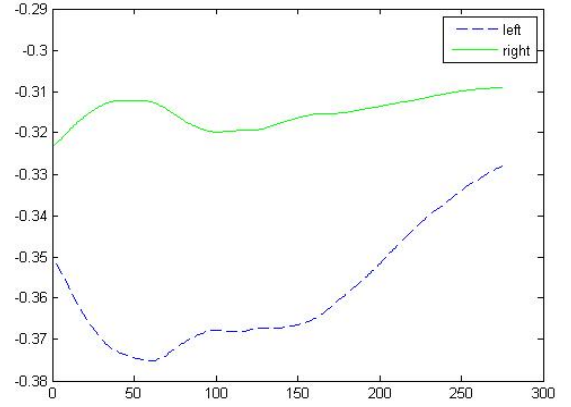


Figure 6. The feature signals of the left-right hands imagery movement

D. Classification

In order to demonstrate the effectiveness of the proposed features in BCI application, the FLDA classification method and SVM method are used in this paper. For the sake of self-containment, this section briefly introduced the FLDA and SVM.

a) FLDA Classifier

The goal in Fisher's linear discriminant analysis (FLDA) is to compute a discriminate vector that separates two or more classes as well as possible. Here we consider the two-class case. We were given a set of input vectors $x_i \in \mathbb{R}^D$, $i \in \{1 \dots N\}$ and the corresponding class-labels $y_i = \{-1, 1\}$. Denoting by N_1 the number of training examples for which $y_i = 1$, by C_1 the set of indices i for which $y_i = 1$, and using analogous definitions for N_2 , C_2 , the objective function for computing a discriminate vector $w \in \mathbb{R}_D$ is

$$J(w) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} \quad (4)$$

$$\mu_k = \frac{1}{N} \sum_{i \in C_k} w^T x_i \quad (5)$$

$$\sigma_k^2 = \sum_{i \in C_k} (w^T x_i - \mu_k)^2 \quad (6)$$

This means that one was searching for discriminate vectors that result in a large distance between the projected means and small variance around the projected means (small within-class variance). To compute directly the optimal discriminate vector for a training data set, matrix equations for the quantities $(\mu_1 - \mu_2)^2$ and $\sigma_1^2 + \sigma_2^2$ can be used. We first define the class means μ_k :

$$m_k = \frac{1}{N} \sum_{i \in C_k} x_i \quad (7)$$

Now we can define the between-class scatter matrix S_B and the within-class scatter matrix S_W .

$$S_B = (m_1 - m_2)(m_1 - m_2)^T \quad (8)$$

$$S_W = \sum_{k=1}^2 \sum_{i \in C_k} (x_i - m_k)(x_i - m_k)^T \quad (9)$$

With the help of these two matrices, the objective function for FLDA can be written as a Rayleigh quotient:

$$J(w) = \frac{w^T S_B w}{w^T S_W w} \quad (10)$$

By computing the derivative of J and setting it to zero, one can show that the optimal solution for w satisfies the following equation:

$$w \propto S_W^{-1} (m_1 - m_2) \quad (11)$$

The classification is conducted as follows:

$$x \in \begin{cases} \text{Class 1} & \text{if } y > 0 \\ \text{Class 2} & \text{if } y < 0 \end{cases} \quad (12)$$

b) SVM Classifier

SVM is a classification algorithm based on statistical learning theory. The invention of SVM is driven by underlying statistical learning theory, i.e., following the principle of structural risk minimization that is rooted in VC dimension theory, which makes its derivation even more profound. The SVM has been a topic of extensive research with wide applications in machine learning and engineering. The output of a binary SVM classifier can be computed by the following expression:

$$f(x) = \text{sgn} \left(\sum_{i=1}^n \alpha_i^* y_i K(x_i, x) + b^* \right) \quad (13)$$

Where $\{x_i, y_i\}$ $N_i=1$ are training samples with input vector $x_i = R_d$; class labels $y_i \in \{-1, 1\}$, $\alpha_i \geq 0$ are Lagrangian multipliers obtained by solving a quadratic optimization problem; b is the bias, and $k\{x_i, x_j\}$ is called kernel function in SVM. Here only use the one typical kernel functions: linear kernel.

$$K(x_i, x_j) = x_i^T x_j \quad (14)$$

where σ is the kernel parameter. It is fair to note that the parameters C and σ should be tuned properly for accurate prediction of unseen samples. In this study, they are selected using LIBSVM by 5-fold cross-validation over the following grid of (C, σ) : $(2^{-7} \dots 2^7) \times (2^{-10} \dots 2^3)$.

The protruding characteristics of SVM lies in its elegant mechanism of handling nonlinear function classes, i.e., nonlinear information processing is carried out by means of linear techniques in an implicit high-dimensional feature space mapped by a nonlinear transformation $\phi(x_i)$ from original input space. Although the analytical expressions of $\phi(x_i)$ is unknown, only the inner product operations $\phi(x_i)^T \phi(x_j)$ are involved, the kernel functions can be used to substitute the inner product operations according to the Mercer theorem. Vapnik's theory show that the SVM solution is found by minimizing both the error on the training set (empirical risk) and the complexity of the hypothesis space, expressed in terms of VC dimension. In this sense, the decision function found by SVM is a tradeoff between learning error and model complexity.

III. RESULTS AND CONCLUSION

Taking into account the different number of samples of experiments may have an impact on classification results, the data of the three subjects are carried out on grouping at the beginning of the experiment. All of the groups are consisted of 250 training samples and 30 test samples, recorded as Subject 1, Subject 2, respectively. Experimental results are showed in Table 1. From the Table 1, we could see that in the case of using Subject 1 (or Subject 2) data set, the results of Subject 1 and Subject 2 are almost the same when we used Fisher linear discriminant analysis method, when we use SVM for classification, the classification results are quite close. Also, we could see that the feature in C3/C4 channel was better than it in P3/P4 channel. In addition the results obtain from SVM method are superior to the results obtain from FLDA method for both algorithms.

TABLE I. CLASSIFICATION OF DIFFERENT METHODS

Discriminate Criterion		DWT+FLDA	DWT+SVM
C3/C4	Subject 1	81.2%	82.1%
	Subject 2	70%	89.7%
P3/P4	Subject 1	79.4%	79.5%
	Subject 2	71.5%	77.1%
Average classification rate		75.5%	82.1%

The above experimental results show that both the FLDA classifier and SVM classifier are effective based on DWT feature extraction when identifying the different mental tasks from EEG signals; but the SVM classifier is more effective.

The experimental results show that the feature extraction and classification algorithm is rational, we also accumulate experience and lay the foundation of the following study of online brain-computer interface system.

We could also see that the classification method is offline, and the subject is normal. In the future, we would make further

study on online BCI for the people with movement disorder, which would promote BCI application in rehabilitation project.

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